

Natanael Detamor Karo Karo

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Summary

Detail-oriented Software Engineer specializing in Laravel and Vue.js with a proven track record of building complex enterprise systems and AI-driven applications. Experienced in architecting end-to-end solutions, from automated KPI engines and resource management workflows to AI inference engines using Python. Passionate about optimizing system performance and delivering high-quality, scalable web applications. Currently seeking a full-time opportunity to leverage technical expertise in a professional engineering team.

Projects

Enterprise Resource Planning (ERP) System – Hopemedia.id

June 2025 - Present

Full-Stack Developer (Laravel, Vue.js, Tailwind CSS, MySQL)

- Architected a multi-phase production workflow supporting 20+ specialized roles with automated status tracking and role-based access control (RBAC).
- Engineered a high-performance HRIS module featuring dynamic leave quota tracking, multi-tier approval hierarchies, and Google Calendar API integration.
- Optimized frontend and backend performance through Lazy Loading, API Caching, and resolving N+1 query issues, significantly increasing response speeds.
- Designed a modern UI/UX using Glassmorphism principles and Tailwind CSS, ensuring a responsive and scalable design system across multiple devices.
- Implemented automated KPI tracking and data visualization (Weekly Charts) to monitor production health and broadcasting performance metrics.

Expert System for Plant Disease Diagnosis – College Research

Nov 2025- April 2026

Full-Stack & AI Developer (Laravel, FastAPI, Python Experta, Vue.js)

- Architected an AI-driven diagnostic engine using Python Experta, implementing Forward Chaining and Certainty Factor (CF) algorithms for high-precision identification.
- Developed a hybrid backend architecture, integrating FastAPI (AI Engine) with Laravel (Business Logic) for seamless real-time data synchronization.
- Implemented a robust security layer featuring OTP (One-Time Password) authentication via WhatsApp and Email for secure user access.
- Engineered a dynamic diagnostic UI in Vue.js with Pinia state management to process complex rule-matching into an intuitive user questionnaire.
- Automated diagnostic reporting by generating downloadable PDF records, allowing users to store and track plant health history.

Education

Universitas Advent Indonesia

August 2022 – June 2026 (Expected)

Bachelor of Informatics Engineering | GPA: 3.61/4.00

Relevant Coursework: Object-Oriented Programming, Data Structures, Software Engineering, Database Management Systems, Web Development I & II, Cloud Computing, Artificial Intelligence, Business Intelligence.

Certifications

- IT Specialist: Databases – Certiport | Mar 2025
- Basic JavaScript Programming – Dicoding | Jan 2025
- Basic Back-End with JavaScript – Dicoding | Jan 2025
- Structured Query Language (SQL) – Dicoding | Apr 2025

Skills and Other Experience

Hard skills: PHP, Golang, JavaScript, Laravel, Vue.js, Node.js, React.js, Tailwind CSS, Bootstrap, MySQL, Git, REST API, JWT, ERP Development, Workflow Automation, Logic Business Design.

Soft skills: Problem Solving, Good communication, Visual thinking, Teamwork, and Technical Documentation.